1 **(a)** 32 to 34 **[B1]**

(b) Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

(c) $120 - 100$ (or $120 -$ their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

[Total Marks: 5]

2 **(a)** $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

(b) 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

(c) Reading from curve at cumulative frequency $= 140$ **[M1]**

46 to 48 **[A1]**

[Total Marks: 7]

3 **(a)** Reading from curve at $m = 220$: 81 to 85 **[M1]**

$100 -$ their reading **[M1]**

15 to 19 **[A1]**

(b) $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

(c) Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

[Total Marks: 9]